CHRIST (Deemed to be University)
Department of Computer Science
Master of Computer Applications

Course: MCA412-2 - Research Methodology

LAB Exercise III –

Defining Research Problem.

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Domain :

Marine Ecological Informatics & Bio-Optical Remote Sensing

Lab 3

Lab Exercise III:
Defining Research Problem.

1. What is Missing or Could Be Improved

A synthesis of recent literature shows that satellite monitoring of the Arabian Sea often treats surface reflectance as a snapshot, without fully connecting it to vertical ocean mixing and atmospheric inputs. Standard chlorophyll-a algorithms can misclassify green Noctiluca scintillans blooms because the organism has distinctive optical features, including reflectance near 531 nm and absorption near 510 nm, that conventional blue-green ratios may not capture well. Satellite remote-sensing reflectance (Rrs) can also lose accuracy at high solar zenith angles above 70°, especially during winter. In addition, long-term studies often rely on single-variable chlorophyll time series, where natural variation and temporal autocorrelation can hide ecological changes that may be more visible in multi-band reflectance data.

2. New Questions That Arise

- Can a narrow-band ratio using Rrs(531) and Rrs(510) distinguish Noctiluca scintillans blooms from diatom populations across satellite images?
- How closely do mixed-layer deepening to around 80 m and changes in the thermocline relate to the timing, location, and intensity of winter algal blooms?
- Can atmospheric dust optical depth (AOD) and changes in dust trajectories help predict occasional blooms in stratified open-ocean waters?
- Can neural-network atmospheric correction (NN-V) improve reflectance retrievals at high solar zenith angles without distorting the blue spectral bands?
- Can multivariate regression across multi-spectral Rrs channels, with autocorrelation correction, detect local Arabian Sea greening trends earlier than single-variable chlorophyll indices?

3. Research Questions

- **RQ1:** How can a bio-optical feature pipeline using the narrow-band Rrs(531)/Rrs(510) ratio distinguish Noctiluca scintillans cell density from typical diatom blooms in MODIS and VIIRS satellite imagery?
- **RQ2:** What statistical relationship exists between vertical mixing conditions (mixed-layer depth, vertical velocity, and sea-surface temperature from Copernicus/GODAS reanalysis) and satellite-observed bloom growth across the northern Arabian Sea?
- **RQ3:** To what extent can multivariate regression across multi-spectral reflectance channels, with serial-autocorrelation correction, reveal ecological shifts earlier than traditional univariate chlorophyll time series?

4. How My Research Could Contribute to Filling These Gaps

My research will develop a computational workflow that combines multi-spectral satellite observations with ocean-atmosphere reanalysis data. Using Level-2 MODIS and VIIRS data, I will examine the 510 nm and 531 nm features to help identify Noctiluca blooms and apply geometric quality checks to reduce noise at high solar zenith angles. I will then combine satellite observations with gridded environmental variables, including mixed-layer depth, sea-surface temperature, vertical velocity, and dust AOD. An autocorrelation-corrected multivariate regression framework will be used to explore bloom patterns and longer-term ecological changes in the northern Indian Ocean.

5. Future Research Directions Suggested by the Authors

- Pandi et al. (2020) highlight the value of 510 nm and 531 nm spectral features and regionally tuned bio-optical algorithms for monitoring dinoflagellate blooms.
- Shi and Wang (2022) recommend combining hydrographic profiles and vertical-velocity information with satellite ocean-colour records to understand the physical drivers of Arabian Sea biomass cycles.
- Kuttippurath et al. (2023) call for linked atmospheric transport and biogeochemical modelling to distinguish dust-fertilisation effects from coastal upwelling.
- Li et al. (2024) support the use of advanced machine-learning atmospheric-correction methods to improve ocean-colour data coverage under high solar zenith angles.
- Cael et al. (2023) suggest moving beyond single-variable chlorophyll indices toward multivariate optical-spectrum analysis and cross-sensor calibration.

6. Two Futuristic Directions

- **Deep Learning Ocean–Atmosphere Surrogate Models:** Develop physics-informed neural networks (PINNs) that use satellite radiances and ocean-mixing information to estimate bloom development and possible hypoxia risk, reducing reliance on repeated external numerical simulations.
- **Autonomous In-Situ Drone–Satellite Early-Warning Systems:** Create satellite edge-computing systems that identify unusual bio-optical bloom signals and help direct autonomous surface vehicles and Bio-Argo floats to collect water-quality measurements before severe bloom impacts occur.

7. Problem Statement

This research aims to address the misclassification of mixotrophic Noctiluca scintillans blooms by conventional satellite chlorophyll algorithms and the limited connection between surface optical observations and vertical ocean mixing. It will develop a narrow-band bio-optical feature-extraction approach using multi-satellite reflectance data and combine the results with physical ocean reanalysis in an autocorrelation-corrected multivariate framework to estimate bloom intensity and examine long-term ecological shifts in the northern Arabian Sea.